Information and Energy Beamforming Optimization for Symbiotic AmBC and SWIPT Networks

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***Abstract*—This letter develops a symbiotic network comprising an ambient backscatter communication network and a simultaneous wireless information and power transfer (SWIPT) network. Specifically, passive tags leverage the primary signal from the full-duplex (FD) base station (BS) for energy harvesting (EH) and data transmission to the FD BS, while delivering more power to the downlink SWIPT users. To enhance the symbiosis benefits and suppress the harms between the two networks, we design uplink zero-forcing reception filters and downlink optimal energy and information beamforming at the FD BS, and optimal power splitting factor at users to maximize the tag sum signal-to-interference-plus-noise ratio, subject to the minimum rate and power requirements of the users, tags' EH requirement and the residual self-interference constraints. Compared to random, maximum ratio transmission, and zero-forcing beamforming, our approach yields significant improvements.**

***Index Terms*—Backscatter communication (BackCom), Symbiotic radio (SR), Full-duplex systems, simultaneous wireless information and power transfer (SWIPT).**

I. Introduction

Symbiotic radio (SR) allows a primary network and a secondary network to operate together to share bandwidth, power, and other resources [1]. For example, smart home sensors (tags) in an Internet-of-Things (IoT) network can be integrated with a primary simultaneous wireless information and power transfer (SWIPT) [2] primary network. The secondary tags form an ambient backscatter communication (AmBC) network [3], [4]. This symbiotic relationship improves overall performance, enhances energy efficiency, and utilizes wireless resources more efficiently. The passive tags can exploit the primary signal for energy harvesting (EH) and data transmission. Moreover, tag signals provide additional power to the SWIPT users, reducing SWIPT transmit power requirements.

On the other hand, cognitive radio aims to boost spectrum utilization by identifying and sharing unused or underutilized frequency bands, primarily through bandwidth sharing [5]. In contrast, SR emphasizes collaboration and coordination between two networks to optimize capacity, energy efficiency, and reliability [1], without a direct connection to the primary network.

***Motivation and Our Contributions**:* The performance gains of an AmBC with a primary SWIPT network (Fig. 1) have never been investigated in the literature. To the best of our knowledge, this is the first paper to do so. The FD base station (BS), with multiple antennas for simultaneous signal transmission and reception, transmits a downlink signal, which carries both information and energy to primary users and acts as the RF source for the passive tags. They send data to the FD BS via backscattering. We aim to maximize the mutual benefits and minimize the harm between the two networks. In particular, to maintain and further enhance the performance of the users and tags in terms of the harvested power and achieved rates, the BS applies optimal downlink energy beamforming (EB) and information beamforming (IB) and uplink zero-forcing (ZF) reception filter while suppressing the residual self-interference (RSI). The main contributions are summarized as follows:

- We introduce a novel SR network compromising an AmBC network of passive tags and a primary SWIPT network and maximize the sum signal-to-interference-plus-noise ratio (SINR) of the tags. Our study is the first paper to formulate this problem and develop a detailed solution.
- The proposed optimization problem, $\mathbf{P}$ in (13), is non-convex and NP-hard, and thus we develop an optimization algorithm based on the alternative optimization (AO) approach, exploiting the semi-definite relaxation (SDR) and second-order cone programming (SOCP) methods. Our algorithm optimizes the IB and EB matrices at the BS, and power-splitting (PS) factors at the users. The algorithm also ensures the minimum rate and power requirements of the primary users and the tags' EH constraints, while constraining the RSI.
- Finally, numerical results show the performance and symbiosis gains. For instance, when the BS is at $10\,\mathrm{dBm}$ and with 16 antennas, our method increases the harvested power and rate of a tag by $15.5\,\%$ and $118.2\,\%$ compared to the maximum ratio transmission (MRT) benchmark.

A. Previous Contributions

An SR network of a SWIPT primary network with an AmBC secondary network has yet to be explored and presents several challenges. They include the design of information and energy beamforming, meeting primary rate requirements, ensuring sufficient EH for SWIPT users, enabling passive tag activation, and determining PS factors for efficient EH and data decoding while ensuring coexistence and multiple access of both networks without interference. Moreover, the use of an FD BS as the primary RF source also creates self-interference (SI). Overcoming these obstacles and achieving optimal performance requires innovative techniques and solutions.

Although these challenges have not been studied for a SWIPT+AmBC configuration, [6]–[10] study AmBC and a

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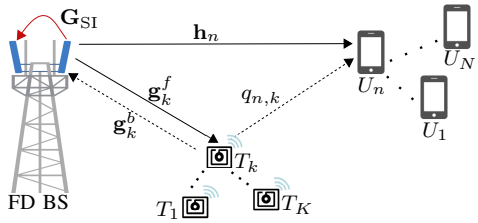

Fig. 1: FD SR setup.

cellular primary network, i.e., no SWIPT. Studies [6], [7] explore time division and non-orthogonal multiple access for EH tags. They, however, assume perfect SI cancellation at the FD BS. Studies [8], [9] consider RSI as noise but do not develop beamforming to suppress RSI. Nonetheless, [10] investigates BS beamforming for a single tag to manage tag interference and minimize BS power while satisfying the tag's rate under RSI. However, the related studies [6]–[9] consider dynamic tags reflection coefficients, which may not be feasible with passive tags. Thus, no prior literature addresses these technical challenges. This letter thus aims to study SWIPT+AmBC networks, addressing the above technical challenges.

Notation: Lower-case bold and upper-case bold denote vectors and matrices. $\mathcal{CN}(\boldsymbol{\mu}, \mathbf{R})$ denotes a complex Gaussian vector with mean $\boldsymbol{\mu}$ and co-variance $\mathbf{R}$. Finally, $\mathcal{K} \triangleq \{1, \ldots, K\}$, $\mathcal{N} \triangleq \{1, \ldots, N\}$, $\mathcal{K}_k = \mathcal{K} \setminus \{k\}$ and $\mathcal{N}_n = \mathcal{N} \setminus \{n\}$.

II. System Model and Preliminaries

A. System and Channel Models

The considered SR system comprises an FD BS (the primary transmitter) with M_t transmit and M_r receive antennas, $N \geq 1$ single-antenna primary users, denoted by U_n $(n \in \mathcal{N})$, and $K \geq 1$ single-antenna tags, T_k $(k \in \mathcal{K})$ - Fig. 1. The BS serves the primary users and simultaneously receives the tag signals. The SWIPT users divide the received power into information decoding (ID) and EH modules [2].

The tags have low processing capabilities and no active RF circuits. The tags also perform EH and data backscattering simultaneously via the PS mode [3]. Therefore, the FD BS transmits a downlink signal which not only carries information to the SWIPT users via IB but also transfers energy to the users and the tags via EB. The FD BS thus enables the tags to backscatter their data to the BS.

We consider block flat-fading channel models. During each fading block, channel response vectors between U_n and the BS is given as $\mathbf{h}_n \in \mathbb{C}^{M_t \times 1}$. Moreover, $\mathbf{g}_k^f \in \mathbb{C}^{M_t \times 1}$ and $\mathbf{g}_k^b \in \mathbb{C}^{M_r \times 1}$ are respectively the forward-link (downlink) and backscattered (uplink) channel vectors between T_k and the BS. And $q_{n,k} \in \mathbb{C}$ is the channel between T_k and U_n. A unified representation of all five channels is given as

$$\mathbf{a} = \zeta_{\mathbf{a}}^{1/2} \tilde{\mathbf{a}}, \tag{1}$$

where $\mathbf{a} \in \{\mathbf{h}_n, \mathbf{g}_k^f, \mathbf{g}_k^b, q_{n,k}\}$, $n \in \mathcal{N}$, $k \in \mathcal{K}$, and $\zeta_{\mathbf{a}}$ accounts for the large-scale path-loss and shadowing. Besides, $\tilde{\mathbf{a}} \sim \mathcal{CN}(\mathbf{0}, \mathbf{I}_M)$ captures Rayleigh fading.

The SI channel, $\mathbf{G}_{\rm SI} \in \mathbb{C}^{M_r \times M_t}$, of the FD BS is modeled as Rician fading channel [11]. The SI channel comprises the line-of-sight (LoS) and the non-line-of-sight (NLoS) parts, which is given by $g_{ij}^{\rm SI} = \sqrt{1/(\bar{K}+1)} g_{ij,\rm LoS}^{\rm SI} + \sqrt{\bar{K}/(\bar{K}+1)} g_{ij,\rm NLoS}^{\rm SI}$, where $g_{ij}^{\rm SI} = [\mathbf{G}_{\rm SI}]_{(i,j)}$, $\bar{K}$ denotes the Rician factor, $g_{ij,\rm LoS}^{\rm SI}$ represent the deterministic part subject to $|g_{ij,\rm LoS}^{\rm SI}|^2 = 1$, and $g_{ij,\rm NLoS}^{\rm SI} \sim \mathcal{CN}(0,1)$. Consequently, higher $\bar{K}$ indicates a strong SI channel at the FD BS.

We thus assume that the FD SR system operates in time-division-duplexing (TDD) mode for channel estimation and data transmission. The system may thus acquire channel state information (CSI) through emerging techniques [12], [13]. We thus assume the availability of perfect CSI.

B. Transmission Model

The signal transmitted by the BS can be expressed as

$$\mathbf{x} = \sum\nolimits_{i \in \mathcal{N}} \mathbf{w}_i s_i + \mathbf{w}_e, \tag{2}$$

where $\mathbf{w}_i \in \mathbb{C}^{M_t \times 1}$ and s_i are the IB weights and the data symbol intended for the i-th primary user, respectively. We assume that s_i's $(i \in \mathcal{N})$ are independent standard circular symmetric complex Gaussian variables, $\mathcal{CN}(0,1)$ [1]. Moreover, $\mathbf{w}_e \in \mathbb{C}^{M_t \times 1}$ is a dedicated EB that enables EH at the primary users and tags. We assume that $\mathbf{w}_e \sim \mathcal{CN}(\mathbf{0}, \mathbf{W}_e)$. The FD BS thus transmits downlink signal $\mathbf{x}$ (2) and simultaneously receives the backscattered data from the tags.

*1) **Tags' Data Transmission & EH**:* T_k $(k \in \mathcal{K})$ receives the signal $(\mathbf{g}_k^f)^{\rm H}\mathbf{x}$ in the downlink and divides the received RF signal power (P_k) into two parts; It reflects αP_k of the received power for data transmission and absorbs the remaining part, i.e., $P_k^{in} = (1-\alpha)P_k$, for EH, where $\alpha \in (0,1)$ is the common reflection coefficient of the tags [4]. Hence the input power to the T_k's EH circuit is given as

$$P_k^{\rm in} = (1-\alpha)\left(\sum\nolimits_{i \in \mathcal{N}} |(\mathbf{g}_k^f)^{\rm H}\mathbf{w}_i|^2 + |(\mathbf{g}_k^f)^{\rm H}\mathbf{w}_e|^2\right). \tag{3}$$

In order to backscatter data, we assume T_k uses a multi-level ($\tilde{M}$-ary) modulation scheme [3]. The tag symbol c_k $(\mathbb{E}\{|c_k|^2\} = 1)$, is thus selected from a $\tilde{M}$-ary constant-envelope modulation. The FD BS thus receives the following signal in the uplink

$$\mathbf{y}^{\rm ul} = \sum\nolimits_{k \in \mathcal{K}} \sqrt{\alpha}\mathbf{g}_k^b(\mathbf{g}_k^f)^{\rm H}\mathbf{x}c_k + \mathbf{G}_{\rm SI}\mathbf{x} + \mathbf{z}^{\rm ul}, \tag{4}$$

where the first term is the backscattered data from the tags, the second term represents the SI, and $\mathbf{z}^{\rm ul} \sim \mathcal{CN}(\mathbf{0}, \sigma_{z^{\rm ul}}^2 \mathbf{I}_{M_r})$, is the additive white Gaussian noise (AWGN) at the FD BS.

In this letter, we utilize ZF beamforming for tag data detection [11], [14]. ZF beamforming effectively reduces inter-tag interference, safeguards against information leakage, and attains the desired rate/detection performance. It closely approaches optimal minimum mean square error (MMSE) beamforming under specific conditions [11]. Future research can investigate alternative schemes or optimal designs.

The FD BS thus applies ZF decoding, i,e., $\mathbf{u}_k = \mathbf{G}_b(\mathbf{G}_b^{\rm H}\mathbf{G}_b)^{-1}\mathbf{e}_k$, to decode T_k's data, where $\mathbf{G}_b = [\mathbf{g}_1^b, \ldots, \mathbf{g}_K^b] \in \mathbb{C}^{M_r \times K}$ and $\mathbf{e}_k$ is the kth column of $\mathbf{I}_K$. It can be easily checked that $\mathbf{u}_k^{\rm H}\mathbf{g}_{k'}^b = 0$ for $k \neq k'$. Hence, the inter-tag interference is suppressed, and the post-processed received signal for T_k will be

$$y_k^{\rm ul} = \sqrt{\alpha}\mathbf{u}_k^{\rm H}\mathbf{g}_k^b(\mathbf{g}_k^f)^{\rm H}\mathbf{x}c_k + \mathbf{u}_k^{\rm H}\mathbf{G}_{\rm SI}\mathbf{x} + \mathbf{u}_k^{\rm H}\mathbf{z}^{\rm ul}. \tag{5}$$

SI in (5) cannot be canceled perfectly due to the limited dynamic range of the receiver even if the SI channel is perfectly known. We thus model the RSI after cancellation at each receive antenna as an independent zero-mean Gaussian distortion noise whose variance is proportional to the received power of the antenna, i.e., $S_{\mathrm{SI},k}, k \in \mathcal{K}$, given as

$$S_{\mathrm{SI},k} = \mathrm{Tr}\Big(\beta \mathbf{U}_k \mathrm{diag}\Big(\mathbf{G}_{\mathrm{SI}}\mathbf{W}_e\mathbf{G}_{\mathrm{SI}}^{\mathrm{H}} + \sum\nolimits_{i\in\mathcal{N}} \mathbf{G}_{\mathrm{SI}}\mathbf{w}_i\mathbf{w}_i^{\mathrm{H}}\mathbf{G}_{\mathrm{SI}}^{\mathrm{H}}\Big)\Big), \quad (6)$$

where $0 < \beta \ll 1$ is a constant denoting the SI cancellation ability of the FD BS, and $\mathbf{U}_k = \mathbf{u}_k\mathbf{u}_k^{\mathrm{H}}$.

*2) **Users' Received Data & EH**:* U_n receives the signal

$$y_n = \mathbf{h}_n^{\mathrm{H}}\mathbf{x} + \sum\nolimits_{k\in\mathcal{K}} \sqrt{\alpha}\mathbf{f}_{n,k}^{\mathrm{H}}\mathbf{x}c_k + z_n, \quad (7)$$

where $\mathbf{f}_{n,k} = q_{n,k}\mathbf{g}_k^f$. In (7), the first term is the direct-link BS signal, the second term is the sum of tag signals, and $z_n \sim \mathcal{CN}(0, \sigma_z^2)$ is the AWGN at U_n.

U_n splits y_n into two parts, where $\rho_n \in (0,1)$ is utilized for ID, and $(1-\rho_n)$ is used for EH, in which ρ_n is the PS factor. Thus, the ID and EH module inputs are given as $y_n^{\mathrm{ID}} = \sqrt{\rho_n}y_n + z_n^{\mathrm{I}}$, and $y_n^{\mathrm{EH}} = \sqrt{(1-\rho_n)}y_n$, where $z_n^{\mathrm{I}} \sim \mathcal{CN}(0, \sigma_{\mathrm{I}}^2)$ is the additional noise at the ID module. Therefore, U_n converts, P_n^{in} (8) into direct current.

*3) **EH Model at Passive Tags & Users**:* The EH circuit of each tag and user harvests the incident RF power P_j^{in}, which is given by P_k^{in} (3) or P_n^{in} (8)[1]. Thus, the harvested power at the j-th device, $P_{h,j}$, can be modeled as a linear or nonlinear function of P_j^{in}. The widely-used linear model has $P_{h,j} = \eta_b P_j^{\mathrm{in}}$, where $\eta_b \in (0,1]$ is the power conversion efficiency. Despite its simplicity, the linear model does not account for saturation and other non-linear effects. Thus, the nonlinear model has $P_{h,j} = \Phi(P_j^{\mathrm{in}})$, where $\Phi(.)$ is a function representing non-linear effects [15, eq. (1)]. Moreover, for both the tags and users, the power harvest must exceed an activation threshold, i.e., $P_{h,j} \geq p_{b,i}$, where $i \in \{u,t\}$ and $p_{b,i}$ is the threshold. The threshold requirement can be expressed in terms of incident RF power as $P_j^{\mathrm{in}} \geq p'_{b,i}$, where $p'_{b,i} \triangleq \Phi^{-1}(p_{b,i})$ and $p'_{b,i} = p_{b,i}/\eta_b$ for non-linear and linear cases. The activation threshold for commercial passive tags and SWIPT users are $-20\,\mathrm{dBm}$ and $-22\,\mathrm{dBm}$ [3], [15].

C. Achievable Rates

We next derive the achieved rates of U_n $(n \in \mathcal{N})$ and T_k $(k \in \mathcal{K})$. Using (7), the achievable rate of U_n is given as

$$\mathcal{R}_n = \log_2(1+\gamma_n), \quad (9)$$

where γ_n is the received SINR at U_n, and is obtained as (10)[2], where it is assumed that $\mathbf{w}_e$ is known to all users [16]. Hence, the interference induced by the energy beam can be eliminated before decoding the desired user's data.

The achieved SINR of T_k, is given as

$$\gamma_k^T = \frac{\alpha|\mathbf{u}_k^{\mathrm{H}}\mathbf{g}_k^b|^2\left(|(\mathbf{g}_k^f)^{\mathrm{H}}\mathbf{w}_e|^2 + |(\mathbf{g}_k^f)^{\mathrm{H}}\mathbf{x}_1|^2\right)}{S_{\mathrm{SI},k} + \|\mathbf{u}_k\|^2\sigma_{z^{\mathrm{ul}}}^2}, \quad (11)$$

[1]EH circuit designs for passive tags and their performance are discussed in [3].

[2]We assume that the tag and the users have the same symbol rate, hence the backscatter-link signals appears as interference to the primary users [1].

where $\mathbf{x}_1 = \sum_{i\in\mathcal{N}} \mathbf{w}_i s_i$ and $S_{\mathrm{SI},k}$ is given in (6). Note that because uplink transmission is powered by the downlink signals, these downlink signals are treated as fast-fading channel gains. Hence, by taking average over the downlink signal s_i $(i \in \mathcal{N})$, R_k is obtained as (Appendix A)

$$R_k = \log_2(e)\Big(\ln(1+\kappa_k) - e^{(1+\kappa_k)\bar{\lambda}_k}\mathrm{E}_i(-(1+\kappa_k)\bar{\lambda}_k)\Big), \quad (12)$$

where $\kappa_k \triangleq a_k\alpha'_k/b_k$, and $\bar{\lambda}_k \triangleq b_k\lambda_k/\alpha'_k$, in which $a_k \triangleq |(\mathbf{g}_k^f)^{\mathrm{H}}\mathbf{w}_e|^2$, $\alpha'_k \triangleq \alpha|\mathbf{u}_k^{\mathrm{H}}\mathbf{g}_k^b|^2$, $b_k \triangleq S_{\mathrm{SI},k} + \|\mathbf{u}_k\|^2\sigma_{z^{\mathrm{ul}}}^2$, and $\lambda_k \triangleq 1/|(\mathbf{g}_k^f)^{\mathrm{H}}\mathbf{W}|^2$, where $\mathbf{W} = [\mathbf{w}_1, \ldots, \mathbf{w}_N] \in \mathbb{C}^{M_t\times N}$. Moreover, $\mathrm{E}_i(x) = \int_{-\infty}^{x} u^{-1}e^u du$.

III. Problem Formulation

We next develop the maximization of the sum SINR of the tags while ensuring tag and primary user (SINR/rate) requirements. We optimize the IB matrix $\mathbf{W}$, EB vector $\mathbf{w}_e$ and the users' PS factors, i.e., $\boldsymbol{\rho} = [\rho_1, \ldots, \rho_N]^{\mathrm{T}}$. We assume a fixed reflection coefficient for passive tags [3]. The optimization problem is thus equivalently formulated as

$$\begin{aligned}
\mathbf{P}: \underset{\mathbf{W},\mathbf{w}_e,\boldsymbol{\rho}}{\text{maximize}} \;& \sum\nolimits_{k\in\mathcal{K}} \gamma_k^T, && (13a)\\
\text{subject to} \;& C_1: \gamma_n \geq \gamma_{\mathrm{th}}, \quad \text{for} \quad n\in\mathcal{N}, && (13b)\\
& C_2: P_n^{in} \geq p'_{b,u}, \quad \text{for} \quad n\in\mathcal{N}, && (13c)\\
& C_3: P_k^{in} \geq p'_{b,t}, \quad \text{for} \quad k\in\mathcal{K}, && (13d)\\
& C_4: S_{\mathrm{SI},k} \leq \delta, \quad \text{for} \quad k\in\mathcal{K}, && (13e)\\
& C_5: \sum\nolimits_{i\in\mathcal{N}} \|\mathbf{w}_i\|_2^2 + \mathrm{Tr}(\mathbf{W}_e) \leq p_t, && (13f)\\
& C_6: \mathbf{W}_e \succeq 0, && (13g)\\
& C_7: 0 < \rho_n < 1, \quad \text{for} \quad n\in\mathcal{N}, && (13h)
\end{aligned}$$

where δ is the maximum allowable RSI at the BS. In $\mathbf{P}$, the constraint (13b) ensures the minimum rate quality of the users, while (13c) and (13d) respectively guarantee the minimum power requirements at the users and tags. Besides, (13f) is the normalization constraint for the BS transmit beamforming. Due to non-convex objective function and constraints, $\mathbf{P}$ is non-convex. Hence, we first split it into two subproblems and solve them using SDR and SOCP approaches.

IV. Proposed Solutions

A. Transmit Beamforming

For a given PS factor $\boldsymbol{\rho}$, $\mathbf{P}$ is reduced to a transmit beamforming optimization problem as follows:

$$\begin{aligned}
\mathbf{P}_w: \underset{\bar{\mathbf{W}},\mathbf{W}_e}{\text{maximize}} \;& \sum\nolimits_{k\in\mathcal{K}} \frac{A_k(\bar{\mathbf{W}},\mathbf{W}_e)}{B_k(\bar{\mathbf{W}},\mathbf{W}_e)}, && (14a)\\
\text{subject to} \;& C'_1: \frac{\mathrm{Tr}\left(\mathbf{H}_n\mathbf{W}_n\right)}{\gamma_{th}} - \sum_{i\in\mathcal{N}_n} \mathrm{Tr}\left(\mathbf{H}_n\mathbf{W}_i\right)\\
& -\sum_{k\in\mathcal{K}}\alpha\left(\sum_{i\in\mathcal{N}}\mathrm{Tr}(\mathbf{F}_{n,k}\mathbf{W}_i) + \mathrm{Tr}(\mathbf{F}_{n,k}\mathbf{W}_e)\right) \geq \sigma_z^2 + \frac{\sigma_{\mathrm{I}}^2}{\rho_n}, && (14b)\\
& C_2 - C_4, && (14c)\\
& C'_5: \sum_{i\in\mathcal{N}} \mathrm{Tr}\left(\mathbf{W}_i\right) + \mathrm{Tr}(\mathbf{W}_e) \leq p_t, && (14d)\\
& C'_6: \mathbf{W}_n, \mathbf{W}_e \succeq 0, && (14e)\\
& C_8: \mathrm{Rank}(\mathbf{W}_n) = 1, \quad \text{for} \quad n\in\mathcal{N}, && (14f)
\end{aligned}$$

$$P_n^{\text{in}} = (1-\rho_n)\left(\sum\nolimits_{i\in\mathcal{N}} |\mathbf{h}_n^{\text{H}}\mathbf{w}_i|^2 + |\mathbf{h}_n^{\text{H}}\mathbf{w}_e|^2 + \sum\nolimits_{k\in\mathcal{K}} \alpha\left(\sum\nolimits_{i\in\mathcal{N}} |\mathbf{f}_{n,k}^{\text{H}}\mathbf{w}_i|^2 + |\mathbf{f}_{n,k}^{\text{H}}\mathbf{w}_e|^2\right)\right) \tag{8}$$

$$\gamma_n = |\mathbf{h}_n^{\text{H}}\mathbf{w}_n|^2 \Big/ \left(\sum\nolimits_{i\in\mathcal{N}_n} |\mathbf{h}_n^{\text{H}}\mathbf{w}_i|^2 + \sum\nolimits_{k\in\mathcal{K}} \alpha\left(\sum\nolimits_{i\in\mathcal{N}} |\mathbf{f}_{n,k}^{\text{H}}\mathbf{w}_i|^2 + |\mathbf{f}_{n,k}^{\text{H}}\mathbf{w}_e|^2\right) + \sigma_z^2 + \sigma_{\text{I}}^2/\rho_n\right) \tag{10}$$

where $\bar{\mathbf{W}} \triangleq \{\mathbf{W}_1, \ldots, \mathbf{W}_N\}$, $\mathbf{W}_n = \mathbf{w}_n\mathbf{w}_n^{\text{H}}$, $\mathbf{H}_n = \mathbf{h}_n\mathbf{h}_n^{\text{H}}$, and $\mathbf{F}_{n,k} = \mathbf{f}_{n,k}\mathbf{f}_{n,k}^{\text{H}}$. Finally, we have

$$A_k(\bar{\mathbf{W}},\mathbf{W}_e) \triangleq \alpha|\mathbf{u}_k^{\text{H}}\mathbf{g}_k^b|^2\left(\sum_{i\in\mathcal{N}} \text{Tr}\left(\mathbf{G}_k^f\mathbf{W}_i\right) + \text{Tr}\left(\mathbf{G}_k^f\mathbf{W}_e\right)\right), \tag{15}$$

$$B_k(\bar{\mathbf{W}},\mathbf{W}_e) \triangleq \beta\left(\sum_{i\in\mathcal{N}} \text{Tr}(\mathbf{Q}_k\mathbf{W}_i) + \text{Tr}(\mathbf{Q}_k\mathbf{W}_e)\right) + \frac{\|\mathbf{u}_k\|^2}{\sigma_{z^{\text{ul}}}^{-2}}, \tag{16}$$

where $\mathbf{G}_k^f = \mathbf{g}_k^f(\mathbf{g}_k^f)^{\text{H}}$ and $\mathbf{Q}_k = \mathbf{G}_{\text{SI}}^{\text{H}}\mathbf{u}_k\mathbf{u}_k^{\text{H}}\mathbf{G}_{\text{SI}}$. We note that $\mathbf{P}_w$ has a non-convex objective function (14a) and rank-one constraint (14f). We apply Dinkelbach's transform. We next reformulate $\mathbf{P}_w$ as the following SDR problem by relaxing the non-convex rank-one constraint:

$$\mathbf{P}_{w1}: \underset{\bar{\mathbf{W}},\mathbf{W}_e}{\text{maximize}} \ \sum\nolimits_{k\in\mathcal{K}} A_k(\bar{\mathbf{W}},\mathbf{W}_e) - \lambda^* B_k(\bar{\mathbf{W}},\mathbf{W}_e), \tag{17a}$$

$$\text{subject to} \ \ C_1', C_2 - C_4, C_5', C_6' \tag{17b}$$

where λ is the auxiliary variable with the optimal value as

$$\lambda^* = \underset{\bar{\mathbf{W}},\mathbf{W}_e}{\text{maximize}} \quad \sum\nolimits_{k\in\mathcal{K}} \frac{A_k(\bar{\mathbf{W}},\mathbf{W}_e)}{B_k(\bar{\mathbf{W}},\mathbf{W}_e)}. \tag{18}$$

For a given λ, $\mathbf{P}_{w1}$ is convex and can be solved by using semi-definite programming (SDP) via CVX Matlab. Next, if the SDR solution satisfies $\text{Rank}(\mathbf{W}_n) = 1$ for $n \in \mathcal{N}$, the optimal solutions are $\mathbf{w}_n = \mathbf{u}_{n,1}$, where $\mathbf{W}_n^* = \mathbf{U}_n\mathbf{S}_n\mathbf{U}_n^{\text{H}}$ and $\mathbf{U}_n = [\mathbf{u}_{n,1} \ldots, \mathbf{u}_{n,M_t}]$. Otherwise, we use Gaussian randomization (GR) to get a near-optimal solution to $\mathbf{P}_{w1}$ [17]. However, satisfying the rank-one constraint (14f) by using GR may cause a slight rate degradation $\mathbf{P}_{w1}$. Algorithm 1 summarizes the steps.

Algorithm 1 : Transmit beamforming.

Initialization: Initialize $\mathbf{w}_n$ and $\mathbf{w}_e$ to feasible values.
Repeat
  Steps 1-3: Update λ by (18); Solve (17); SVD $\mathbf{W}_n^* = \mathbf{U}_n\mathbf{S}_n\mathbf{U}_n^{\text{H}}$.
  if $\text{Rank}(\mathbf{W}_n^*) = 1, \forall n$, **then**
    return: $\mathbf{w}_n^* = \mathbf{u}_{n,1}$ for $n \in \mathcal{N}$.
  else
    for $d = 1, \ldots, D$ **do**
      1: $\mathbf{v}_{n,d} = \mathbf{U}_n\mathbf{S}_n^{1/2}[\text{e}^{j\phi_1}, \text{e}^{j\phi_2}, \ldots, \text{e}^{j\phi_{M_t}}]^{\text{H}}$.
      2: Check if $\mathbf{P}_{w1}$ is feasible with $\mathbf{v}_{n,d}$ for $n \in \mathcal{N}$.
    end for
    return: $\mathbf{w}_n^* = \mathbf{v}_n^*$, where $\mathbf{v}_n^* = \underset{d=1,\ldots,D}{\arg\min}\ \mathbf{v}_{n,d}$.
  end if
Until the value of the objective function converges.
Output: Optimal transmit precoders $\mathbf{w}_n^*$ for $n \in \mathcal{N}$.

B. Power Splitting

For given $\mathbf{w}_n$ and $\mathbf{w}_e$, we get PS/power allocation problem:

$$\mathbf{P}_\rho: \text{find} \ \ \boldsymbol{\rho}, \tag{19a}$$

$$\text{subject to} \ \ C_1, C_2, C_7. \tag{19b}$$

As this resembles relay beamforming for the multi-antenna relay broadcast channel [18], we transform it into a SOCP problem and solve via CVX Matlab [18].

TABLE I: Simulation settings

Parameter	Value	Parameter	Value
B	10 MHz	δ	$p_t/10$
N_f	10 dB	M_t, M_r	16,16
σ_I^2	-80 dBm	N, K	2, 3
β	-90 dB	$d_{h_n}, n \in \mathcal{N}$	12 m
$\bar{K}$	3 dB	$d_{g_k}, k \in \mathcal{K}$	$\mathcal{U}(5,6)$m
α	0.6	$d_{q_{n,k}}, n \in \mathcal{N}, k \in \mathcal{K}$	7 m
$\rho_n, n \in \mathcal{N}$	0.6	$p_{b,u}, p_{b,t}$	-22 dBm

Note that the computational complexities of the SDR and SOCP methods have been extensively studied. Additionally, since problem $\mathbf{P}$ is NP-hard and non-convex, estimating the gap between the proposed solution and the global optimal solution appears intractable.

V. Simulation Results

Table I gives the simulation parameters. The fading model is described in [3, *Sec. IV.C.6*]. Moreover, the AWGN variance is set at $\sigma_{z^{\text{ul}}}^2 = \sigma_z^2 = 10\log_{10}(N_0BN_f)$ dBm, where $N_0 = -174$ dBm/Hz, B is the bandwidth, and N_f is the noise figure. The minimum user rate is set to 2 bps/Hz. We label three benchmarks (with equal power allocation) and the proposed scheme as follows:

- A_1 : *Random-Random*: $\mathbf{w}_n$ and $\mathbf{w}_e$ are normalized complex Gaussian random vectors.
- A_2 : *MRT-Random*: $\mathbf{w}_n = \mathbf{h}_n/\|\mathbf{h}_n\|$ and $\mathbf{w}_e$ is a normalized complex Gaussian random vector.
- A_3 : *ZF-Random*: $\mathbf{w}_n = \mathbf{H}(\mathbf{H}^{\text{H}}\mathbf{H})^{-1}\mathbf{e}_n/\|\mathbf{H}(\mathbf{H}^{\text{H}}\mathbf{H})^{-1}\mathbf{e}_n\|$, where $\mathbf{H} = [\mathbf{h}_1, \ldots, \mathbf{h}_N]$, and $\mathbf{w}_e$ is a normalized complex Gaussian random vector.
- A_4 : *Proposed*: $\mathbf{w}_n$, $\mathbf{w}_e$, and ρ_n are the solutions to (13).

For A_1-A_3, we set $\rho_n = 0.6$ for $n \in \mathcal{N}$. We must note that random beamforming does not require CSI, and the beam can be focused in any direction, resulting in poor performance. MRT, on the other hand, directs the beam toward the SWIPT users and thus necessitates accurate CSI. Unlike MRT, ZF cancels out the interference between the SWIPT users.

Fig. 2 illustrates the power delivered to the users and tags relative to the number of BS antennas for p_t at 10 dBm. It is evident that our algorithm excels in delivering substantial power to both the users and the tags, surpassing the activation threshold, even with limited resource consumption at the BS (e.g., a limited number of antennas and low transmit power). In contrast, benchmarks A_1, A_2, and A_3 necessitate an excessive number of antennas to maintain the required power level for the tags. Furthermore, with A_1, the received power at the users is significantly insufficient to reach the activation threshold.

Fig. 3 and Fig. 4 respectively depict the user sum rate and the tag sum rate under different BS beamforming designs. We see that A_3, based on ZF beamforming, achieves the highest sum rate for the SWIPT users, while A_1 with random beamforming, results in poor rates. Furthermore, MRT achieves a sum rate close to the proposed design. These three, however,

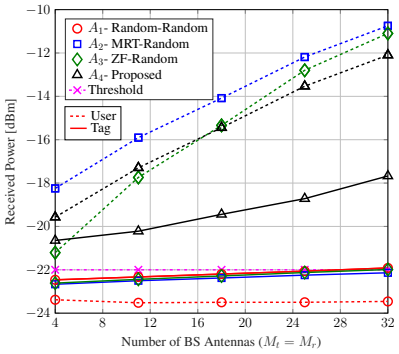

Fig. 2: User/tag received power for $p_t = 10\,\mathrm{dBm}$.

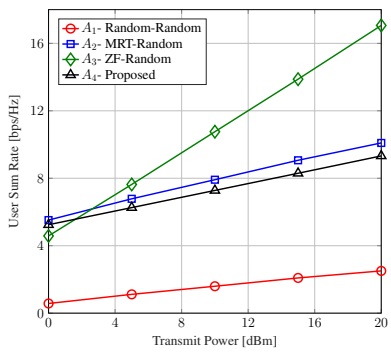

Fig. 3: User sum rate as a function of transmit power.

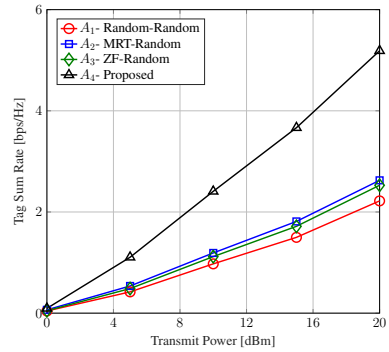

Fig. 4: Tag sum rate as a function of transmit power.

achieve poor tag sum rates (Fig. 4). This is because, with MRT and ZF, the BS solely focuses on the users and ignores the tags, while random beamforming may or may not achieve a gain toward the tags. Besides, RSI at the BS, which is ignored in A_1, A_2, and A_3, can further degrade the tag rates. In contrast, our algorithm improves the mutual benefits and suppresses the inter-network interference between, and also mitigates the RSI at the FD BS. It not only improves the harvested power at the tags and the users (Fig. 2) but also achieves acceptable performance for the users, and enhances the tag sum rate significantly. In particular, at $p_t = 10\,\mathrm{dBm}$, compared to ZF, although the user sum rate degrades by $48\,\%$, our scheme improves the tag sum rate by $137\,\%$ and harvested power by $15\,\%$, while achieving required user rates.

VI. Conclusion

This paper developed an SR network of an AmBC network and a SWIPT network. To maximize the symbiotic gains, encompassing achievable rates and harvested power, while effectively minimizing the negative effects such as interference, we optimized EB and IB at the BS, as well as PS factors at the SWIPT users. By leveraging a combination of AO, SDR, SOCP, and Dinkelbach's methods, we effectively tackled the non-convex nature of the problem. Our algorithm successfully maximized the overall sum rate of the tags while simultaneously ensuring satisfactory performance for SWIPT users and guaranteeing the minimum power requirements for the tags. Through this simultaneous improvement in terms of harvested power and achieved rates, our proposed algorithm promotes a constructive coexistence between the two networks, facilitating their mutual benefits and enhancing overall system performance.

Appendix A
Average Tag's Rate (12)

To take average over $s_i (i \in \mathcal{N})$, γ_k^T (11) is written as $\gamma_k^T = (\alpha_k'/b_k)(a_k + \mathbf{s}^{\mathrm{H}}\mathbf{W}^{\mathrm{H}}\mathbf{g}_k^f(\mathbf{g}_k^f)^{\mathrm{H}}\mathbf{W}\mathbf{s})$, where $a_k \triangleq |(\mathbf{g}_k^f)^{\mathrm{H}}\mathbf{w}_e|^2$, $b_k \triangleq S_{\mathrm{SI},k} + \|\mathbf{u}_k\|^2\sigma_{z^{\mathrm{u1}}}^2$, $\alpha_k' \triangleq \alpha|\mathbf{u}_k^{\mathrm{H}}\mathbf{g}_k^b|^2$, and $\mathbf{s} = [s_1, \ldots, s_N]^{\mathrm{T}} \in \mathbb{C}^{N\times 1}$. We note that $z_r \triangleq (\mathbf{g}_k^f)^{\mathrm{H}}\mathbf{W}\mathbf{s}$ is $\mathcal{CN}(0, |(\mathbf{g}_k^f)^{\mathrm{H}}\mathbf{W}|^2)$. Thus $z_a \triangleq |z_r|^2$ is exponential with $\lambda_k = 1/|(\mathbf{g}_k^f)^{\mathrm{H}}\mathbf{W}|^2$. The PDF of γ_k^T is thus a truncated exponential given by $f_{\gamma_k^T}(\gamma) = e^{\lambda_k a_k}\bar{\lambda}_k e^{-\bar{\lambda}_k\gamma}$, for $\gamma \geq a_k\alpha_k'/b_k$, where $\bar{\lambda}_k \triangleq b_k\lambda_k/\alpha_k'$. And averaging $\ln(1+\gamma_k^T)$ over this PDF, we get (12).

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